3D Coupled Modelling of a Series-Connected Bifilar Cu-Stabilised REBCO Resistive SFCL

H-formulation, solid heat transfer and external circuit solved together on a 50 mm representative segment of four tape runs, 400 V source model

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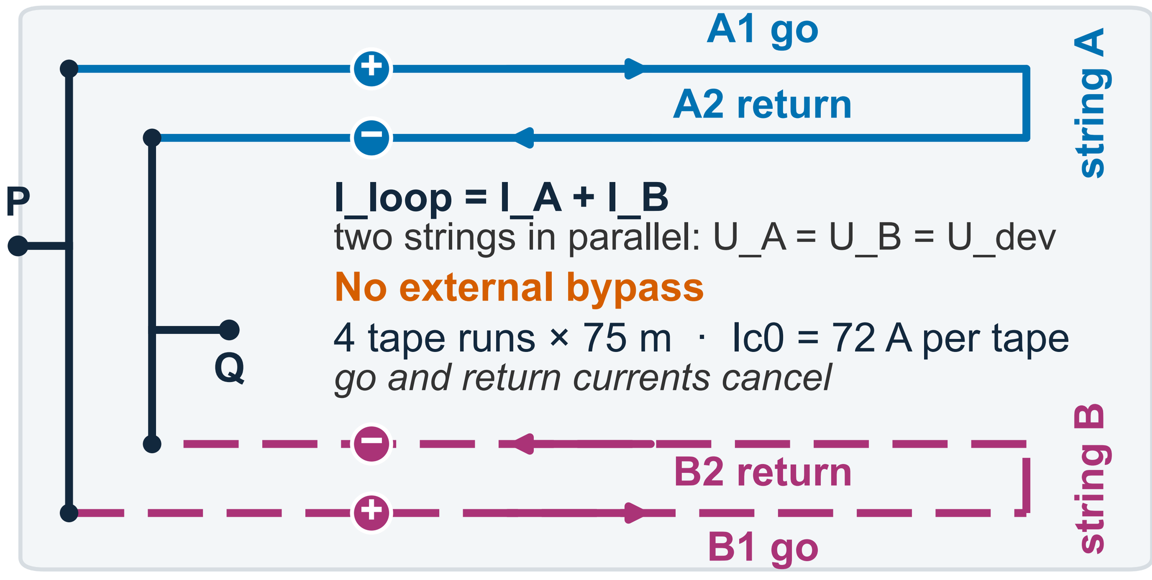
84.2 %
independent peak current reduction
(the two peaks are 3.6 ms apart)

79.37 K
volume mean peak in the 20 ms window;
local extremum 79.67 K, mesh sensitive

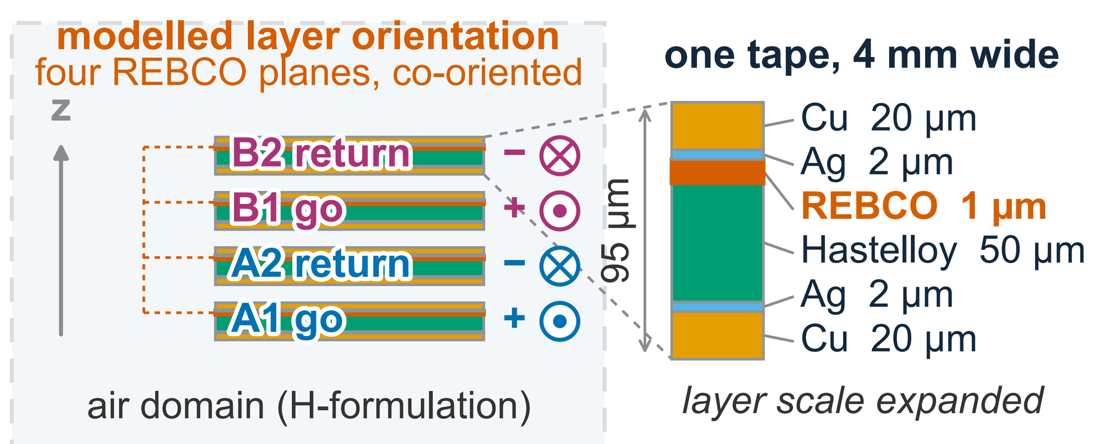
309.3 V
voltage peak across the device,
0.4 ms after the current peak

1 Device topology and length scaling

(a) BCS electrical topology

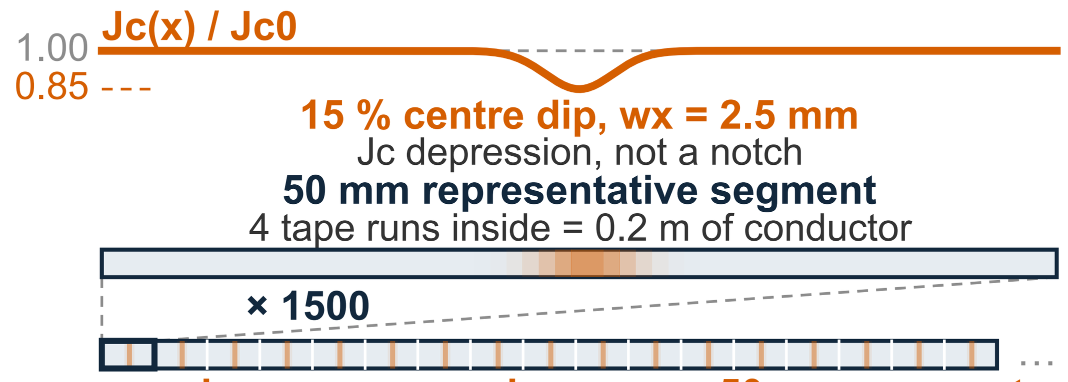


(b) Stacked go–return pairs and conductor layer order



current along x, out of the page

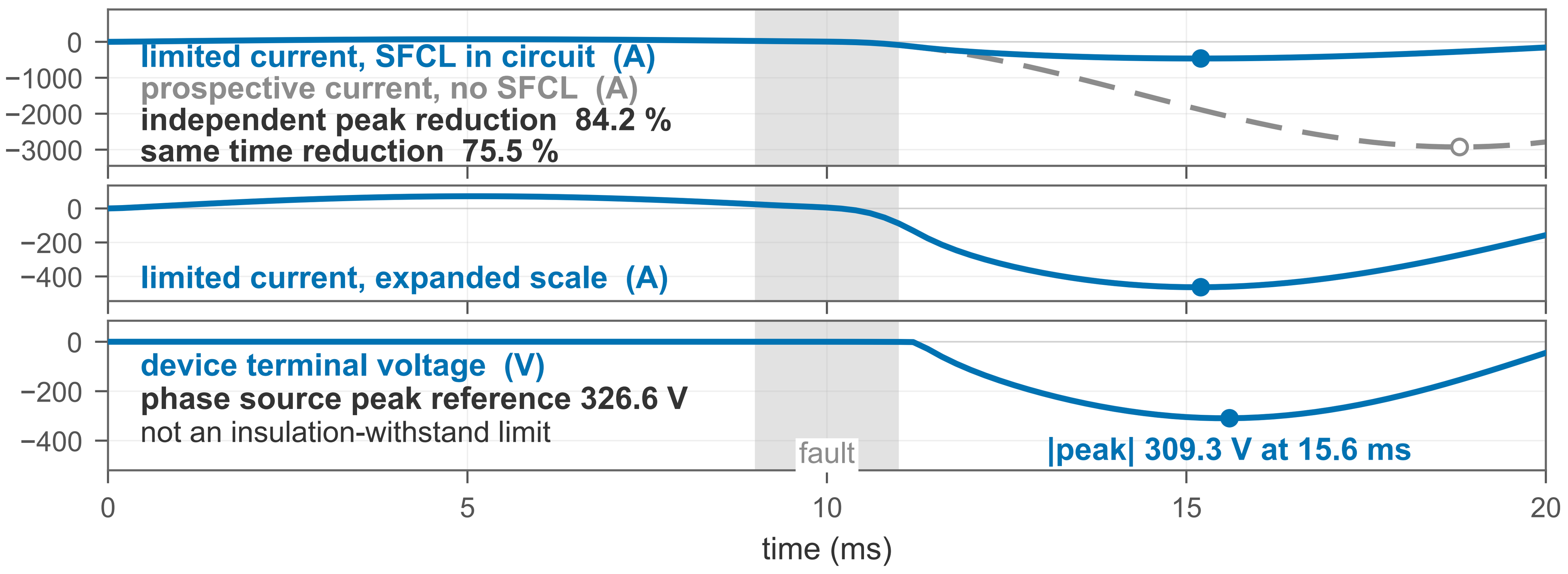
(c) Segment → assumed array → device



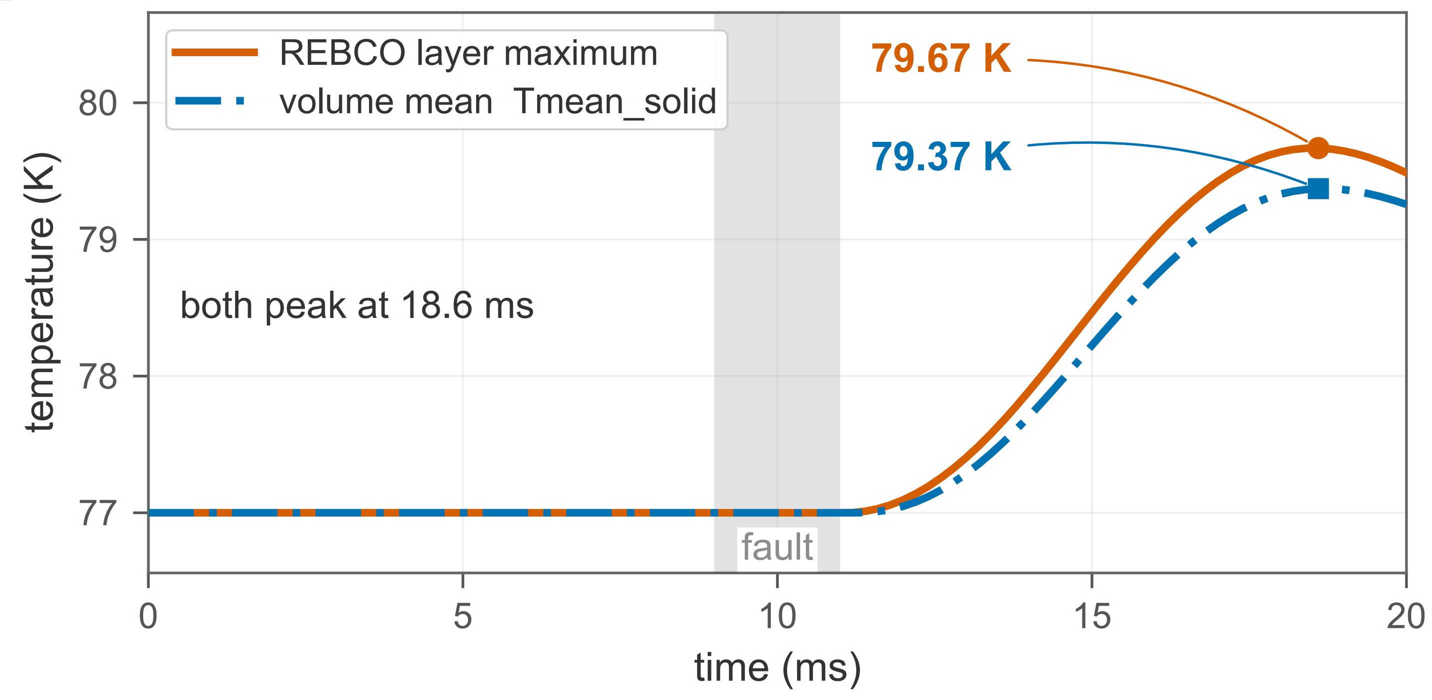
assumed array: one weak zone per 50 mm segment
150 m per string
two strings: 300 m total tape inventory
representative segment + length scaling

Two hairpin strings in parallel; in each, outgoing and return tapes are in series so their currents cancel. Panel **(b)** is the four stacked tapes, panel **(c)** the modelled segment and its length scaling.

2 Circuit and thermal response



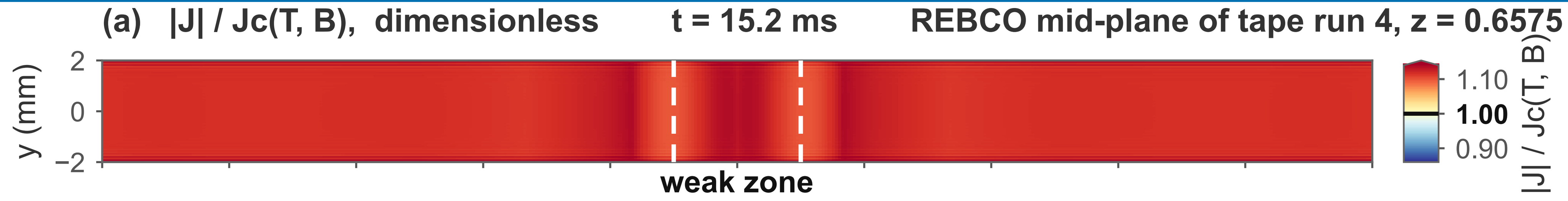
Current limiting and device terminal voltage on one shared time axis. The voltage peak lags the current peak, so the two are not one operating point; the modelled peaks are negative and the sign carries no design meaning.



Temperature, double reported: the volume mean over the 24 solid domains, and the extremum, a maximum over volume value on the 1 μm REBCO layer that is never a design point value on its own.

3 Spatial electromagnetic and thermal fields

(a) |J| / Jc(T, B), dimensionless t = 15.2 ms REBCO mid-plane of tape run 4, z = 0.6575 mm

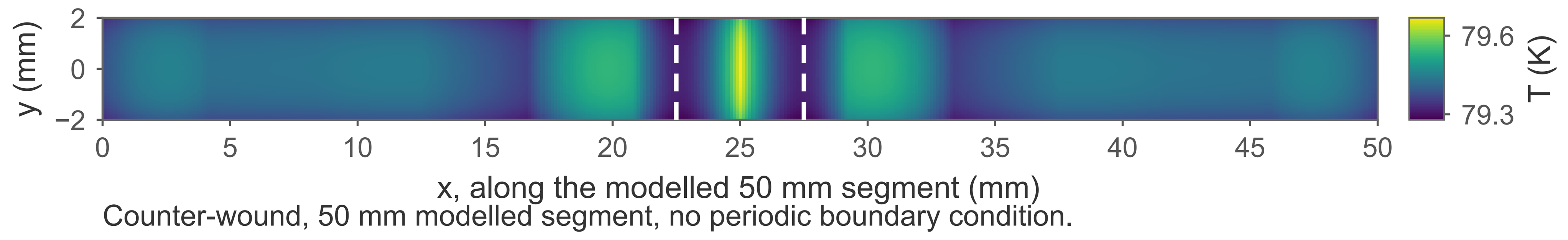


|J| is a curl derived field; the tape edge band and the position of its maximum are both set by the mesh.

the whole layer is above unity at this instant: 100.00 % of the sampled plane
weak zone: centre x = 25 mm, wx = 2.5 mm, minimum Jc/Jc0 = 0.85
(a) plane maximum is a sampled lower bound on the extremum over the domain

plane max 1.2309
on a tape edge row, |y| = 2.00 mm; x position set by the mesh

(b) T (K) t = 18.6 ms same plane, same grid, same view as (a)



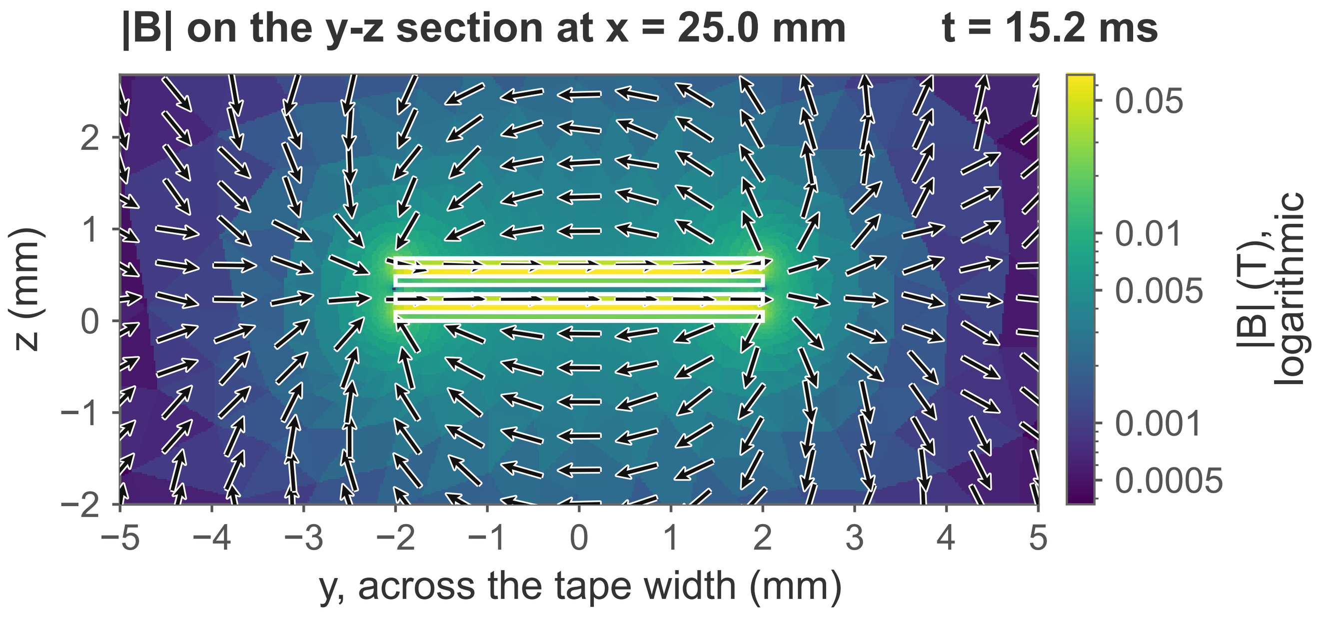
Counter-wound, 50 mm modelled segment, no periodic boundary condition.

plane max (grid-sampled) 79.66796806 K · mean 79.41655240 K
min 79.27894020 K
Reference-mesh field; hotspot magnitude remains mesh-sensitive.

plane maximum at x = 25.00 mm, y = 0.00 mm

The same plane, same grid, at the two instants the figure names: normalised current density above, temperature below. Dashed lines mark the weak zone, and both panels share one axial axis.

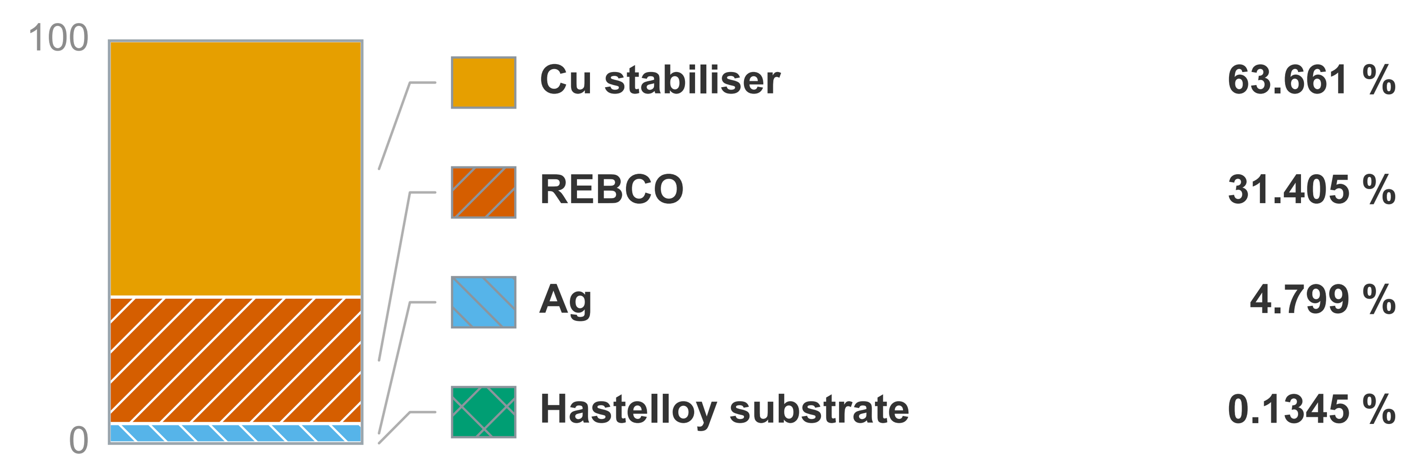
|B| on the y-z section at x = 25.0 mm t = 15.2 ms



Tape run current directions
⊙ current out of page (+x)
⊗ current into page (−x)
⊙ run 4
⊗ run 3
⊙ run 2
⊗ run 1
sampled maximum |B|: 0.0681 T
→ direction only, uniform length

Direction symbols read from the exported field, by Ampère circulation around each tape run.

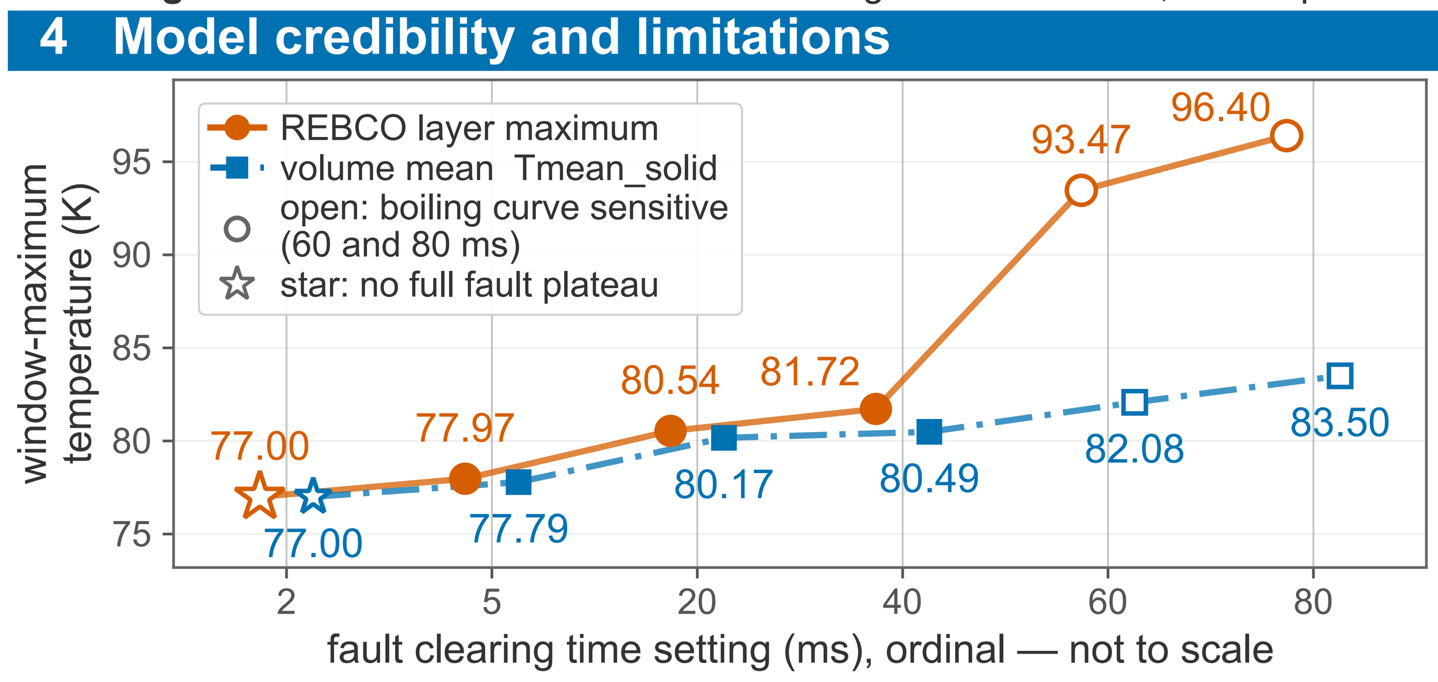
share of window E·J energy (%)



Share of the modelled solid E · J energy over the window, by material group. It is a segment-level time integral, not an instantaneous power split, and length scaling does not change it.

Field magnitude on the transverse section through the weak zone, with in-plane direction arrows.

4 Model credibility and limitations



Six discrete breaker settings, one run each. The calibres differ between settings, so the six are not one homogeneous scan, nothing between them is interpolated, and the two longest remain unresolved.

Model scope.
- Coupled 3D H-formulation, solid heat transfer and circuit in one unsegmented 20 ms window: 10 ms energising, then 10 ms faulted.
- Two parallel strings, counter-wound to suppress net inductance; no external bypass.
- A 50 mm representative segment containing four tape runs, discretised with 77,928 elements at a relative tolerance of 10^{-4}; device values follow by length scaling.

Outstanding limitations.
- Axial mesh refinement has not entered an asymptotic range, and the thickness and transverse directions were never solved, so no error bar is quoted.
- The 60 ms and 80 ms clearing settings remain unresolved, being sensitive to the boiling curve superheat axis.
- The 0.1 mm inter-tape spacing is assumed, as is one weak zone per 50 mm segment.
- Nothing here has been calibrated against an independent formulation.

Take home. Within this model line the device holds the line current far below the prospective fault current, the solid stays within about three kelvin of the bath, and most of the window E · J energy goes to the copper stabiliser.

References
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[2] B. Shen et al., Supercond. Sci. Technol. 33 (2020) 033002. doi:10.1088/1361-6668/ab66e8
[3] N. Riva et al., Supercond. Sci. Technol. 33 (2020) 114008. doi:10.1088/1361-6668/aba34e
[4] B. V. Balakin et al., Cryogenics 65 (2015) 5–9. doi:10.1016/j.cryogenics.2014.11.003
Solver settings, repeats and core counts: see the appendix.

yuumkaaax.github.io/sfcl